For Examiner's Use

**2** A hot-air balloon rises vertically at a constant speed. At time $t$ = 0, a ball is released from the balloon.
Fig. 2.1 shows the variation with time $t$ of the ball's velocity $v$.

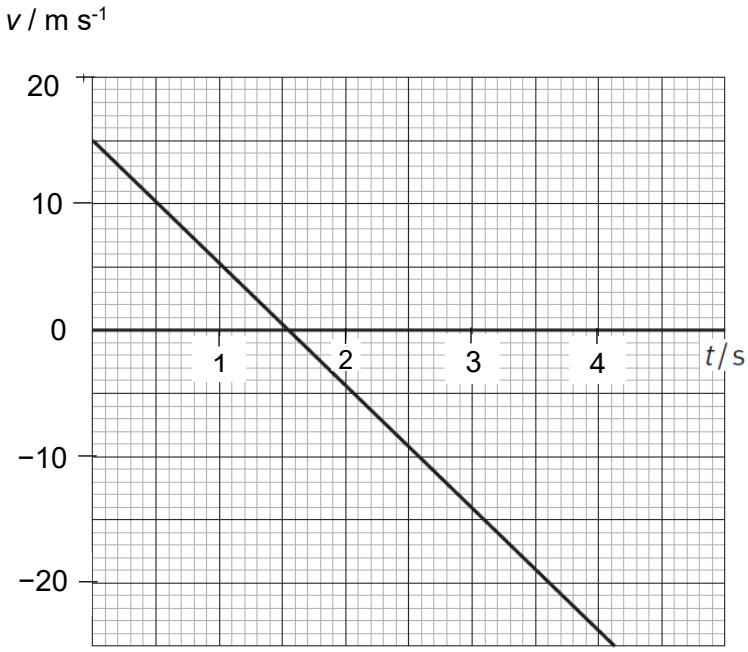


**Fig. 2.1**

The ball hits the ground at $t$ = 4.1 s.

**(a)** State the speed of the hot-air balloon.

speed = ..... m s$^{-1}$ [1]

**(b)** Explain how the graph in Fig. 2.1 shows that the acceleration of the ball is constant.

.....

.....[1]

**(c)** Use Fig. 2.1 to

**(i)** state the time at which the ball reaches its highest point,

time = ..... s [1]

**(ii)** show that the ball rises for a further 12 m between release and its highest point,

[1]

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**(iii)** determine the distance between the point of release of the ball and the ground.

distance = ..... m [2]

**(d)** Describe the difference between displacement of the ball and the distance it travels.

.....

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.....[2]

**(e)** Sketch a new graph on Fig. 2.1 showing the variation with *t* of the ball's velocity *v* if air resistance is not negligible. Assume terminal velocity is attained by the ball before hitting the ground. Label the new graph N. [2]